

for all e-governance projects. This brings in a level playing field for Open Source software in the multi-billion dollar e-governance market in India.

Similarly, private organisations are realising the importance of migrating to OSS. Sectors like BFSI and telecom currently derive the maximum benefit of OSS. They are realising that OSS eventually yields better results in terms of quality, performance and pricing.

Adoption challenges in enterprises

The adoption of OSS at the desktop in enterprises faces several challenges because of the legacy of poorly written client-server applications, and Web applications 'optimised' for Internet Explorer 5.0, etc. Some of the other key hurdles are discussed below:

Applications—the biggest bottleneck for Linux adoption

Some time ago, a major financial institution in India was evaluating Linux as an option for its desktops. It had more than 50 custom applications, most of which were Web applications that had been developed over the last decade. The Web applications were supposedly platform-agnostic, but when they were tested on a Linux desktop, it was discovered that just 10 per cent of the applications worked flawlessly. The rest had many compatibility issues, especially with Active-X controls, IE-specific Java script code, the use of IE-specific code, invocation of locally installed Windows applications from within the Web applications, etc. The company found that the only way it could successfully migrate to Linux on the desktop was by first making all its applications truly platform agnostic at the client level.

An interesting side effect of migration is that many organisations suddenly realise the importance of having access to the source code. When an organisation wants to transform its legacy apps to cross-platform apps, it often finds that it does not have access to the source code to most of the custom developed applications. It is almost impossible to make any changes to the applications, as the firms do not have any ongoing maintenance contract with the original vendor that developed the software, nor do they have access to the source code.

Major ISVs are not enthusiastic

This is a classic chicken-or-egg scenario. The packaged software supplied by major ISVs poses a big challenge in any large scale migration initiative. Independent Software Vendors (ISVs) claim that they would support Linux better if there is demand from the market. Enterprises insist that they would migrate when there is better support for Linux from the ISVs.

Organisational practices

When a large media organisation in India decided to migrate to Linux as the OS platform, the biggest hindrance it faced was that it had dozens of MS-Office (MS-Excel) macros that had been developed and, in most cases, its partner organisations also built their workflows around these macros. Now, for migration, the organisation had to make an inventory of all its macros, document and rationalise their use, and get them translated into Open Office Macros with a significant investment of money and time, before it could adopt Linux and Open Office.

Solutions and opportunities

These hurdles can be easily overcome with a six-step approach. Here are the six essential steps to open source migration:

- Discovery
- Analysis
- Design
- Planning
- Implementation
- Maintenance

Quality time and effort spent in the Discovery and Analysis phase, will lead to an optimal solution design. Implementation itself is generally done in multiple phases for an easy roll out. The trick for successful implementation is to prioritise the applications and attack those that are most compatible with the system, and then deal with the non-compatible applications.

Enterprises can exercise freedom of choice on the OS front by taking the determined stand 'If it is not cross-platform, it is not in', and embarking on a drive to make current applications platform agnostic. This will not only make the Linux migration more effective and sustainable, but will also have the additional benefit of being ready for any future devices.

To summarise, there is an increased awareness about the benefits of the open source model and a realisation that these benefits extend far beyond the cost advantages. There is also a demand for tools and products developed under a truly open source model supported by enterprises and government. 

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